

1st Part Problem Set for Applied Abstract Algebra

Quivers: representations and morphisms

- Let $M, M' \in \text{rep } Q$. Show that the set of morphism $\text{Hom}(M, M')$ is a k -vector space.
- Let Q be the quiver $1 \longrightarrow 2 \longleftarrow 3$ and consider the representations

$$M \quad k \xrightarrow{\begin{bmatrix} 1 \\ 0 \end{bmatrix}} k^2 \xleftarrow{\begin{bmatrix} 0 \\ 1 \end{bmatrix}} k$$

$$M' \quad k \xrightarrow{\begin{bmatrix} 1 \\ 0 \end{bmatrix}} k^2 \xleftarrow{\begin{bmatrix} 1 \\ 0 \end{bmatrix}} k$$

- Show that M and M' are not indecomposable.
 - Show that M and M' are not isomorphic.
- Let Q be the quiver $1 \longrightarrow 2 \longleftarrow 3$ and let M be a representation

$$k^2 \xrightarrow{\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}} k^3 \xleftarrow{\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}} k .$$

- Write M as a direct sum of the indecomposable representations listed below:

$$S(1) : k \longrightarrow 0 \longleftarrow 0 \quad S(2) : 0 \longrightarrow k \longleftarrow 0 \quad S(3) : 0 \longrightarrow 0 \longleftarrow k$$

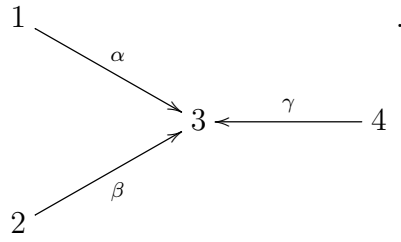
$$P(1) : k \xrightarrow{1} k \longleftarrow 0 \quad I(2) : k \xrightarrow{1} k \xleftarrow{1} k \quad P(3) : 0 \longrightarrow k \xleftarrow{1} k$$

- Show that there is a non-split short exact sequence

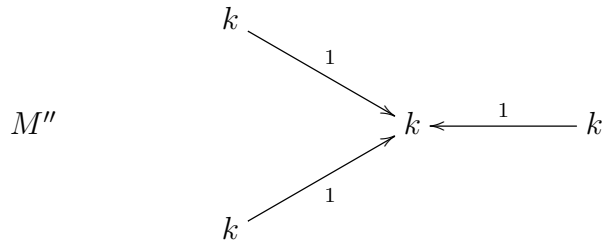
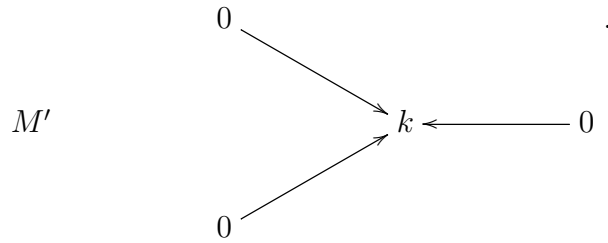
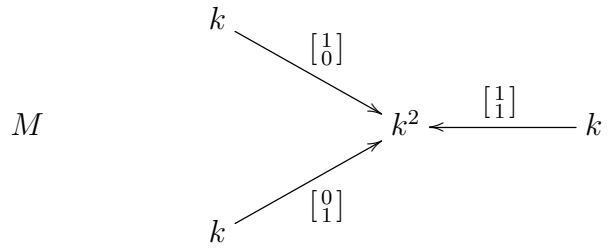
$$0 \longrightarrow X \longrightarrow Y \longrightarrow Z \longrightarrow 0 .$$

- Find all indecomposable representations up to isomorphism of the quiver $1 \longrightarrow 2$.

5. Let Q be the quiver



and consider the following representations



(a) Show that

$$\begin{aligned} \operatorname{Hom}(M, M') &= 0, & \operatorname{Hom}(M', M) &\simeq k^2 \\ \operatorname{Hom}(M, M'') &\simeq k^2, & \operatorname{Hom}(M'', M) &= 0. \end{aligned}$$

(b) Show that M is not isomorphic to $M' \oplus M''$.

(c) Show that there is a short exact sequence:

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0.$$

6. Let Q be the quiver

$$1 \begin{array}{c} \xleftarrow{\alpha} \\ \xleftarrow{\beta} \end{array} 2$$

For any $\lambda \in k \cup \{\infty\}$, define M_λ to be the representation:

$$1 \begin{array}{c} \xleftarrow{1} \\ \xleftarrow{\lambda} \end{array} 2 \quad \lambda \in k;$$

$$1 \begin{array}{c} \xleftarrow{0} \\ \xleftarrow{1} \end{array} 2 \quad \lambda = \infty.$$

- (a) Show that each M_λ is indecomposable.
- (b) Show that $M_\lambda \simeq M_\mu$ if and only if $\lambda = \mu$. In particular, the number of indecomposable representations depends on the choice of the field k .
- (c) Show that $\text{Hom}(M_\lambda, M_\mu) = 0$ if $\lambda \neq \mu$.
- (d) Show that for each λ there is a short exact sequence

$$0 \longrightarrow S(1) \longrightarrow M_\lambda \longrightarrow S(2) \longrightarrow 0 ,$$

where 1 and 2 are the representations

$$S(1) : k \begin{array}{c} \xleftarrow{\quad} \\ \xleftarrow{\quad} \end{array} 0 \quad S(2) : 0 \begin{array}{c} \xleftarrow{\quad} \\ \xleftarrow{\quad} \end{array} k .$$

- 7. Let $f : L \rightarrow M$ be a morphism of representations. Prove that the definition of $\text{coker } f$ is equivalent to the one defined in classroom through universal property (otherwise, see Remark 1.6 in Schiffler's book).
- 8. Let M , M' and N be representations of Q and let $f : M \rightarrow N$, $g : M' \rightarrow N$ be morphisms. Define the fiber product (or pull back) of f and g as

$$X = \{(a, b) \mid a \in M, b \in M', \text{ such that } f(a) = g(b)\},$$

and define the projections $\pi_1(a, b) = a$ and $\pi_2(a, b) = b$.

- (a) Show that X is a subrepresentation of $M \oplus M'$ and that there is a commutative diagram:

$$\begin{array}{ccc} X & \xrightarrow{\pi_2} & M' \\ \pi_1 \downarrow & & \downarrow g \\ M & \xrightarrow{f} & N \end{array}$$

- (b) Show that if f is injective then π_2 is injective.
- (c) Show that if f is surjective then π_2 is surjective.

- (d) Suppose $0 \longrightarrow L \xrightarrow{h} M \xrightarrow{f} N \longrightarrow 0$ is a short exact sequence, define $h' : L \rightarrow X$ by $h'(n) = (h(n), 0)$. Show that the following diagram is commutative with exact rows:

$$\begin{array}{ccccccc}
 0 & \longrightarrow & L & \xrightarrow{h'} & X & \xrightarrow{\pi_2} & M' \longrightarrow 0 \\
 & & \downarrow 1_L & & \downarrow \pi_1 & & \downarrow g \\
 0 & \longrightarrow & L & \xrightarrow{h} & M & \xrightarrow{f} & N \longrightarrow 0
 \end{array}$$